MILITARY SPECIFICATION

FIN AND CLOSURE ASSEMBLY FOR ROCKET MOTOR 2.75 INCH, SR 105, PARTS FOR

1. SCOPE

1.1 This specification covers the quality assurance provisions and special requirements not covered by the drawings, for one type of fin and closure assembly designated as Fin and Closure Assembly For, Rocket Motor 2.75 Inch, SR 105.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-M-232	- Methanol (Methyl Alcohol)
PPP-B-630	- Box Fiberboard
TT-B-846	- Butyl Alcohol; Normal (Butanol) (for Use in Organic Coatings)
TT-I-735	- Isopropyl Alcohol, Technical

MILITARY

MIL-A-2550	- Ammunition and Special Weapons General Specification For.
MIL-I-6868	- Inspection Process, Magnetic Particles
MIL-I-45607	- Inspection Equipment, Supply and Maintenance Of.

STANDARDS

FEDERAL

Federal Test Method STD No. 151.

FSC 1340

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables For Inspection By Attributes.

DRAWINGS

9266959 - Fin and Closure Assembly

PUBLICATIONS

ET 9266959 - Equipment Tabulation

(Copies of specifications, standards drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. - Materials and parts shall be in accordance with drawings and specifications.

3.2 Fin and Closure Assembly (dwg. 9266959). - The fin and closure assembly shall comply with all requirements specified on drawing 9266959 and with all requirements specified in specifications.

3.3 Metal Parts and Other Material. - The metal parts and other material used shall be free from defects, including all surface imperfections and discontinuities and all internal defects in the finished component resulting from flaws or lack of soundness in the materials used.

3.3.1 Surface Imperfections. - Surface imperfections include pits, seams, laps, tears, cracks, scale, blisters, cold shuts, and other defects of similar type.

3.3.2 Internal Defects. - Internal defects include pipes, laminations, segregations, internal cracks, porosity, flakes, inclusions, and other defects of similar type.

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3.4 Magnetic Particle Inspection. - When the Nozzle Plate is subjected to magnetic particle inspection, the following requirements shall be met:

3.4.1 Inclusions of sulphides or other ferrous compounds precipitated and rolled into the material, of any degree in the thick section between the nozzle holes are not acceptable and the part shall be rejected. Shallow discontinuities in other areas of the surface of these holes are acceptable where they are backed up by a 2/16 inch minimum thickness of sound material (See Figure 1).

3.4.2 Inclusions of sulphides or other ferrous compounds precipitated and rolled into the material on either the inside or outside surface of the piston cylinder are unacceptable (See Figure 1).

3.4.3 Inclusions of sulphides or other ferrous compounds precipitated and rolled into the material in the skirt that show on both the inside and outside surfaces are not acceptable and the part shall be rejected. Shallow discontinuities in either surface that do not show on the other surface are acceptable (See Figure 1 and 2).

3.4.4 Cracks or cracklike indications are not allowed except in the following areas:

a. On the surfaces adjacent to the detent groove not to exceed 1/16" in depth.

b. On the outside surface of the skirt that does not extend down into the grooves more than 1/16". No cracks are allowed on the forward face surface of the nozzle plate. No cracks are allowed at the lockwire hole.

3.5 Pivot Pin Load Test. - The pivot pin shall withstand a static load of twenty pounds applied successively to each end with no relative movement between the pivot pin and the fin blade.

3.6 Insulation Integrity. - The Fin and Closure Assembly prior to installation of Shorting Clip must measure ten (10) megohms minimum resistance when measured with one hundred (100) volts between the Contact Disc and the bourrolet on the Nozzle Plate.

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3.7 Lead Pull Strength. - The fin and closure assembly shall be capable of withstanding a twelve (12) pound pull for two (2) seconds between the Nozzle Plate and the forward end of the live lead and between the nozzle plate, (with the Cross Head restrained from rearward movement), and the Contact Disc with the Fin Retainer disengaged from the Fin Blades.

3.8 Grounding Clip Assembly Pull Strength. - The Fin and Closure Assembly prior to installation of the Electrical Lead Live Assembly shall withstand a pull load of two (2) pounds for two (2) seconds applied between the forward end of wire of the soldered ground clip assembly and the closure assembly.

3.9 Piston Pressurization Strength. - The Piston Assembly shall be capable of withstanding a pressure of two thousand (2000) pounds per square inch gage (psig), plus fifty (50) psig, minus zero (0) psig applied to that portion of the assembly forward of the "O" ring groove in the piston. Each Piston Assembly shall be capable of withstanding the above pressure applied in the above manner for two (2) seconds without pressure drop or without leak either at the end of the Piston between the Piston body and the insulation or between the insulation and the conductor.

3.10 Fin Blade Pin Pull. - Each fin blade shall be capable of withstanding a pull test of 2650 pounds between a pin of 0.1886 - .0002" diameter and the aft end of the blade.

3.11 Fin Blade Free Fall. - All Fin Blades with pins assembled on each Fin and Closure Assembly shall be capable of free movement and fall of their own weight when raised 60 degrees with the horizontal.

3.12 Closure Assembly Alcohol Inspection. - All exposed surfaces of the molding compound on the Closure Assembly when subjected to a Alcohol Inspection, shall show no evidence of cracks, voids, inclusions, delaminations, or porous areas.

3.13 Ground lead continuity. - The fin and closure assembly shall not exceed a resistance of 0.10 ohms between the nozzle plate and the forward end of the ground lead.

3.14 Air Tightness. - The closure assembly shall withstand without a leakage rate of 10-4 c.c./sec. or greater an air pressure of 10 psig applied to the inner part of the assembly.

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3.15 Workmanship. - All parts shall be fabricated and finished in a thorough workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, metal shavings, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing bare material. All required markings shall be neat and sharply defined. Known item defects (to include complete compliance with 3.1, 3.2 and 3.3) shall not be submitted with the lot.

3.16 First Article Inspection. - This specification makes provisions for First Article Inspection. Requirements for the submission of First Article Samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product. - At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

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The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Sample

4.2.1 Submission. - The contractor shall submit a First Article Sample to a government approved facility as designated by the Contracting Officer (see para. 6.4) for evaluation in accordance with the provisions of 4.2.2. The First Article sample shall consist of 30 assemblies and 30 complete sets of parts (i.e., 30 each of every component, and every subassembly, plus 10 Fin Blades with pins inserted, and 3 tensile specimens machined from fin blades) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings, certificates of conformance for materials (see 4.1.1). A sample containing known defects will not be submitted unless specifically authorized, by the Contracting Officer. A First Article Sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspection to be Performed. - Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in this specification and any or all requirements of the drawings.

4.2.2.1 Fin Blade Tensile. - The First Article Sample of Fin Blade tensile specimens as supplied (4.2.1) and fin blades with pins inserted, (4.2.1) shall be subjected to both tests as specified in 4.3.3.1.1 and 4.3.3.1.2.

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4.2.3 Rejection. - If any assembly or component fails to comply with any of the requirements and inspections, the First Article Sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and/or submit a new First Article Sample or portion thereof. Until a First Article Sample is accepted, the contractor is, in no way authorized by the Government to initiate regular production unless directed by the Contracting Officer.

4.3 Inspection Provisions for Production.

4.3.1 Lot Formation. - The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of products from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall bear a unique lot number. Unless otherwise approved by the Contracting Officer, the inspection lot size of the item being procured and deliverable under the contract shall be 5000 plus 50 minus 0 Fin and Closure Assemblies.

4.3.2 Examination. - The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that

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continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply the AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspection listed shall be in accordance with 4.3.4.

4.3.2.1 "O" Ring (see dwg. 650953 covering a detail of dwg. 9266959).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Nicks, cuts, surface defects or surface irregularities.....	Visual	01001
102.	I. D.....	Gage	01002
Minor:	AQL 1.00 percent		
201.	Diameter of cross section.....	Gage	01003
202.	Evidence of poor workmanship (see 3.15).....	Visual	01004

4.3.2.2 "O" Ring (see dwg. 650950 covering a detail of dwg. 9266959).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Nicks, cuts, surface defects or surface irregularities.....	Visual	02001
102.	Diameter of cross section.....	Gage	02002
103.	I. D.....	Gage	02003

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Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.15).....	Visual	02004

4.3.2.3 Fin and Closure Assembly (see dwg. 9266959).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Assembly does not pass through launch tube simulator.....	Gage	03001
Major:	AQL 1.0 percent		
101.	Part missing.....	Visual	03002
102.	Fin retainer not securely assembled.....	Visual/ Manual	03003
103.	Ground clip placement improper.....	Visual	03004
104.	Fin spacing unsymetrical.....	Visual	03005
105.	Rubber compound inadequately applied.....	Visual	03006
106.	Contact Disc, connector and lead not properly insulated...	Visual	03007
Minor:	AQL 1.00 percent		
201.	Marking improper or missing...	Visual	03008
202.	Evidence of poor workmanship (see 3.15).....	Visual	03009

4.3.2.4 Nozzle plate after plating - (see dwg. 9266954 covering a detail of 9266959)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 2.5 percent		
101.	Concentricity "O" ring groove diameter to outside diameter at forward section of plate...	Gage	04001

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102.	Concentricity of lockwire groove diameter to outside diameter at forward section of plate	Gage	04002
103.	Concentricity of bourrelet to outside diameter at forward section of plate	Gage	04003
104.	Concentricity of piston hole diameter at aft surface to outside diameter at forward section of plate	Gage	04004
105.	Perpendicularity of lockwire groove to axis of outside diameter at forward section of plate	Gage	04005
106.	Nicks, radial scratches, gouges, and spiral tool marks in "O" ring groove	Visual	04006
107.	Outside diameter of forward section of plate	Gage	04007
108.	"O" Ring groove diameter	Gage	04008
109.	Distance, fin pivot hole centerline, to opposite and adjacent fin pivot holes centerline	Gage	04009
110.	Symmetry, fin pivot holes, to aft O.D	Gage	04010
111.	Surface finish piston hole	Gage	04011
112.	Depth of nozzle port	Gage	04012
113.	Nozzle port counterbore diameter	Gage	04013
114.	Depth of nozzle port counterbore	Gage	04014
115.	Diameter of fin pivot holes	Gage	04015
116.	Parallelism, fin pivot holes, the aft outside diameter, and the forward face of the lockwire groove	Gage	04016
117.	Width of fin pivot slots	Gage	04017
118.	Symmetry, fin pivot slots, to the aft O.D	Gage	04018
119.	Thickness of material below lockwire hole	Gage	04019
120.	Distance from center of plate to inside surface of fin pivot ears	Gage	04020

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121.	Width of lockwire groove.....Gage	04021
122.	Diameter of lockwire groove...Gage	04022
123.	Forward piston hole diameter..Gage	04023
124.	Concentricity, forward piston hole diameter to aft piston hole diameter.....Gage	04024
125.	Diameter of piston hole in aft face of plate.....Gage	04025
126.	Outside diameter at aft part of plate.....Gage	04026
127.	Concentricity, aft outside diameter to outside diameter at forward section of plate...Gage	04027
128.	Inside diameter at forward part of plate.....Gage	04028
129.	Concentricity, inside diameter at forward part of plate to outside diameter at forward part of plate.....Gage	04029
130.	Distance, forward face at lockwire groove to forward face at step, min.....Gage	04030
131.	Distance, spotface on aft section of plate to forward face at piston boss.....Gage	04031
132.	Distance, spotface to fin pivot hole.....Gage	04032
133.	Distance, aft face of plate to forward face of plate.....Gage	04033
134.	Outside diameter of piston boss.....Gage	04034
135.	Distance, forward face of plate to forward face of "O" Ring groove.....Gage	04035
136.	Distance, fin pivot hole centerline to aft face of plate.....Gage	04036
137.	Diameter of bourrelet.....Gage	04037
138.	Concentricity of piston boss to aft outside diameter of plate.....Gage	04038
139.	Location of nozzle port holes.....Gage	04039

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Minor:	AQL 4.00 percent		
201.	Symmetry of lockwire hole to lockwire slot	Gage	04040
202.	Diameter of lockwire hole	Gage	04041
203.	Length from aft face of fin pivot ear to front surface of plate	Gage	04042
204.	Diameter of detent groove	Gage	04043
205.	Diameter of spotface	Gage	04044
206.	Distance, aft piston hole to spotface	Gage	04045
207.	"O" ring groove parallel to forward face of detent groove	Gage	04046
208.	Distance aft face of plate to aft face of detent groove	Gage	04047
209.	Distance, forward face at step to forward face at detent groove	Gage	04048
210.	Concentricity, of detent groove to outside diameter at forward section of plate	Gage	04049
211.	Width of "O" Ring groove	Gage	04050
212.	Evidence of poor workmanship (see 3.15)	Visual	04051

4.3.2.5 Fin Blade (see dwg. 9266951 covering a detail of dwg. 9266959).

Categories	Defects	Method of Inspection	Code No.
Critical:	*None defined.*		
Major:	AQL 1.5 percent		
101.	Blade straightness (Note 9A)	Gage	05001
102.	Twist (Note 9B)	Gage	05002
103.	Blade symmetry (Note 9C)	Visual	05003
104.	Thickness of blade at hole, boss area	Gage	05004
105.	Distance, aft edge of fin to pin hole centerline	Gage	05005
106.	Wall thickness around hole, min	Gage	05006
107.	Surface finish at hub	Gage	05007

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108.	Parallelism, fin blade flat face at pivot pin hole to a plane offset (see dwg. 9266951 Section A-A)	Gage	05008
109.	Diameter of fin pivot hole	Gage	05009
110.	Cross Head bearing face not coincident with fin pivot hole	Gage	05010
111.	Centerline of fin pivot hole perpendicular to plane offset at flat face of fin pivot hole and in true position with cross head bearing face	Gage	05011
112.	Distance, leading edge to cross head bearing corner	Gage	05012
113.	Distance leading edge to center of hole	Gage	05013
Minor:	AQL 2.5 percent		
201.	Height (trailing edge to leading edge)	Gage	05014
202.	Cross head bearing corner radius improper	Gage	05015
203.	Thickness at slotted end	Gage	05016
204.	Protective finish missing or improper	Visual	05017
205.	Radius around hole	Gage	05018
206.	Evidence of poor workmanship (see 3.15)	Visual	05019

4.3.2.6 Plate and Sleeve Assembly (see dwg. 9266955 covering a detail of dwg. 9266959).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	True position of sleeve O.D. to O.D. at forward part of plate	Gage	06001
102.	Nozzle sleeves loose	Visual/ Manual	06002

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103.	Symmetry sleeves to piston hole	Gage	06003
Minor:	AQL 0.65 percent		
201.	Distance, nozzle insert O.D. wall to adjacent nozzle insert O.D. wall, min	Gage	06004
202.	Evidence of poor workmanship (see 3.15)	Visual	06005

4.3.2.7 Pin, Straight, Headless (MS16556-641 covering a detail of dwg. 9266959).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Outside diameter, overall	Gage	07001
102.	Overall length	Gage	07002
Minor:	AQL 1.0 percent		
201.	Evidence of poor workmanship (see 3.15)	Visual	07003
202.	Protective finish missing or improper	Visual	07004

4.3.2.8 Cross Head (see dwg. 456909 covering a detail of dwg. 9266959).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Both faces not perpendicular to piston hole	Gage	08001
102.	Concentricity of piston hole to outside diameter	Gage	08002
103.	Protective finish missing or improper	Visual	08003
Minor:	AQL 1.5 percent		
201.	Diameter of cross head, max	Gage	08004
202.	Thickness	Gage	08005
203.	Width of cross bars	Gage	08006
204.	Evidence of poor workmanship (see 3.15)	Visual	08007

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4.3.2.9 Contact Disc (see dwg. 1253129 covering a detail of dwg. 9266959).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Plating missing or inadequate.	Visual	09001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.15).....	Visual	09002

4.3.2.10 Fin Retainer (see dwg. 1253131 covering a detail of dwg. 9266959).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.15).....	Visual	10001

4.3.2.11 Insert Throat (see dwg. 9266957 covering a detail of dwg. 9266959).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Grain orientation improper....	Visual	11001
102.	Length.....	Gage	11002
103.	Evidence of cracks, voids, inclusions or porous areas....	Visual	11003
Minor:	AQL 2.50 percent		
201.	Outside diameter.....	Gage	11004
202.	Inside diameter.....	Gage	11005
203.	True position of O.D. to I.D..	Gage	11006
204.	Diameter at taper.....	Gage	11007
205.	Evidence of poor workmanship (see 3.15).....	Visual	11008

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4.3.2.12 Sleeve, Exit Cone (see dwg. 9266956 covering a detail of dwg. 9266959).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Outside diameter, aft.....	Gage	12001
102.	Sleeve length.....	Gage	12002
103.	Distance, aft face of sleeve to forward edge of step on outside surface.....	Gage	12003
104.	True position, aft O.D. to forward O.D.....	Gage	12004
105.	Forward I.D.....	Gage	12005
Minor:	AQL 1.00 percent		
201.	Plating missing or improper...	Visual	12006
202.	Evidence of poor workmanship (see 3.15).....	Visual	12007

4.3.2.13 Washer, Expansion (see dwg. 9266958 covering a detail of dwg. 9266959).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined		
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.15).....	Visual	13001

4.3.2.14 Piston (Glass Bead, Two Ends), (see dwg.9220783 covering a detail of 9266959).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 1.00 percent		
101.	Diameter of Knurled section after knurling	Gage	14001
102.	Head diameter	Gage	14002
103.	Body diameter	Gage	14003
104.	Cross head shaft diameter	Gage	14004
105.	Thread improper	Gage	14005
106.	Concentricity of head diameter with body diameter	Gage	14006
Minor:	AQL 1.5 percent		
201.	Length of knurl	Gage	14007
202.	Length of body	Gage	14008
203.	Evidence of poor workmanship (see 3.15)	Visual	14009

4.3.2.15 Piston Assembly (Glass Bead, Two Ends), (see dwg. 9266964 covering a detail of dwg. 9266963).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Glass bead cracked, loose or missing	Visual/ Manual	15001
102.	Piston Plating missing or improper	Visual	15002
103.	Forward wire length	Gage	15003
Minor:	AQL 1.00 percent		
201.	Wire plating missing or improper	Visual/Gage	15004
202.	Evidence of poor workmanship (see 3.15)	Visual	15005

4.3.2.16 Closure Assembly (see dwg. 9266953 covering a detail of dwg. 9266959).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 2.50 percent	
101.	Outside diameter of molded surface.....Gage	16001
102.	Concentricity of forward step of molded surface to outside diameter of metal nozzle plate.....Gage	16002
103.	Forward step outside diameter of molded surface.....Gage	16003
104.	Runout of forward step inside diameter at the forward face of the closure assembly to forward outside diameter of the metal nozzle plate.....Gage	16004
105.	Forward step inside diameter at the forward face of the closure assembly.....Gage	16005
106.	Concentricity of the inside diameter at the aft edge angled molded surface to the outside diameter of the metal nozzle plate.....Gage	16006
107.	Inside diameter at the aft edge of the angled molded surface.....Gage	16007
108.	Diameter of molded cavity around the piston boss.....Gage	16008
109.	Concentricity of the diameter of the molded cavity around the piston boss to the forward outside diameter of the metal nozzle plate.....Gage	16009
110.	Distance, forward face of the closure assembly to the forward face of the lockwire groove.....Gage	16010
111.	Distance forward face of the molded step at forward part of the closure assembly to the forward face of the lockwire groove.....Gage	16011

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112.	Distance, aft face of nozzle to forward edge of nozzle throat venturi.....Gage	16012
113.	Nozzle throat diameter.....Gage	16013
114.	Concentricity, nozzle throat diameter to aft exit cone on molded surface.....Gage	16014
115.	Distance, aft face of nozzle to aft datum point of exit cone on molded surface.....Gage	16015
116.	Distance, aft face of nozzle to forward datum point of exit cone on molded surface...Gage	16016
117.	True position of nozzle exit cone to forward O.D. of nozzle plate.....Gage	16017
118.	Distance, forward face of the molded surface of the closure assembly to the forward edge entrance of the nozzle venturi.....Gage	16018
119.	Nozzle entrance venturi diameter.....Gage	16019
120.	True position, nozzle entrance venturi diameter to forward O.D. of nozzle plate.....Gage	16020

Minor: AQL 2.50 percent

201.	Distance, forward face of the molded surface of the closure assembly to aft face of the molded cavity around the piston boss.....Gage	16021
202.	Distance, forward face of the molded surface of the closure assembly to the forward edge face of molded cavity around the piston boss.....Gage	16022
203.	Distance between forward venturi cavities, min.....Gage	16023
204.	Diameter of the forward molded venturi cavities.....Gage	16024
205.	Angle of forward molded cavity to forward molded face.Gage	16025

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206.	Aft surface of sleeve and molded material flush	Gage	16026
207.	Concentricity of forward overall outside diameter of molded surface to outside diameter of metal nozzle plate	Gage	16027
208.	Evidence of poor workmanship (see 3.15)	Visual	16028

4.3.2.17 Clip (see dwg. 9266961 covering a detail of dwg. 9266959).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Aft I.D.	Gage	17001
102.	Overall length	Gage	17002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.15)	Visual	17003

4.3.2.18 Clip Ass'y Grounding (see dwg. 9266962 covering a detail of dwg. 9266959).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.00 percent		
201.	Overall length	Gage	18001
202.	Evidence of poor workmanship (see 3.15)	Visual	18002

4.3.2.19 Electrical Lead Live Assembly (see dwg. 9266963 coverirg a detail of dwg. 9266959).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 0.65 percent		
101.	Insulation length, max.....	Gage	19001
102.	Bare wire exposed.....	Visual	19002
103.	Insulation loose.....	Visual/ Manual	19003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.15).....	Visual	19004

4.3.2.20 Fin Retainer and Live Lead Assembly (see dwg. 9234433 covering a detail of 9266959).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Overall length.....	Gage	20001
102.	Insulation improper.....	Visual	20002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.15).....	Visual	20003

4.3.2.21 Nut, Self Locking (see dwg. MS 21044-N3 covering a detail of 9266959).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined		
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.15).....	Visual	21001

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4.3.3 Testing

4.3.3.1 Fin Blade Tensile Test-Major Defect. - Each furnace (heat treat) lot of fin blades shall be sampled and tested for fin blade strength. Furnace (heat treat) lots shall be sampled and tested according to the following:

a. According to 4.3.3.1.1, until five consecutive lots have passed fin blade material verification. Thereafter, one lot in five furnace (heat treat) lots shall be randomly selected, and so tested.

b. All furnace (heat treat) lots which are not sampled and tested according to 4.3.3.1.1 shall be sampled and tested according to 4.3.3.1.2.

c. Rejection of any furance (heat treat) lot shall require return to testing as in a. above.

4.3.3.1.1 Fin Blade Material Verification Test-Major Defect. Code No. 05020 (see dwg. 9266951 covering a detail of 9266959). - Three (3) fin blades from each furnace (heat treat) lot to be tested shall be subjected to a tensile pull test as specified in 4.4.1.1 using equipment in accordance with 4.3.4. Failure of two or more fin blades to meet the requirements of dwg. 9266951 shall cause furnace (heat treat) lot rejection. In the event that lot rejection criteria are reached due to no test (machine failure, specimen failing outside gage length, etc.) each no test fin blade shall be replaced in the sample by another fin blade drawn from the heat treat lot.

4.3.3.1.2 Fin Blade Pin Pull Test-Major Defect Code No. 05021 (see 3.10). Ten (10) fin blades from each furnace (heat treat) lot to be tested shall be subjected to a tensile pull test as specified in 4.4.1.2 using equipment in accordance with 4.3.4. Failure of one or more fin blades to meet the requirements of para. 3.10 shall cause furnace (heat treat) lot rejection. Failure of the fin blade to be pulled to the minimum load attributable to the test equipment (such as pin failure) shall be considered a no test fin blade. In the event of more than one no test fin blade in the sample, additional fin blades shall be selected so that nine fin blades shall be validly tested.

4.3.3.2 Compression Test on Cross Head (see dwg. 456909 covering a detail of 9266959)-Major Defect - Code No. 08008. This paragraph shall apply when a powdered metal Cross Head is supplied in the Fin and Closure Assembly. A sample consisting of fifty (50) Cross Heads shall be taken from each lot of Cross Heads.

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The sample shall be tested as specified in 4.4.2 using equipment in accordance with 4.3.4. Failure of any Cross Head in the sample to meet the requirements of Note 8.A. on dwg. 456909 shall cause rejection of the lot.

4.3.3.3 Bending Test on Cross Head (see dwg. 456909 covering a detail of 9266959) - Major Defect - Code No. 08009. This paragraph shall apply when a powdered metal Cross Head is supplied in the Fin and Closure Assembly. A sample consisting of fifty (50) Cross Heads shall be taken from each lot of Cross Heads. The sample shall be tested as specified in 4.4.3 using equipment in accordance with 4.3.4. Failure of any Cross Head in the sample to meet the requirements of Note 8.B. on dwg. 456909 shall cause rejection of the lot.

4.3.3.4 Pressure Test on Piston Assembly (see dwg. 9266964 covering a detail of 9266959) - Major Defect - Code No. 15006. - The sampling plan for this test shall be in accordance with Standard MIL-STD-105 using inspection Level II with an AQL of 0.25 percent. This test shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4. Any Piston Assembly which does not meet the requirements specified in 3.9 shall be considered defective.

4.3.3.5 Magnetic Particle Inspection of Nozzle Plate (see 3.4) - Critical Defect - Code No. 04055. - Each lot of Nozzle Plates before plating shall be subjected to one hundred percent magnetic particle inspection as specified in 4.4.5 using equipment in accordance with 4.3.4. Any Nozzle plate failing to meet the requirements specified in 3.4 shall be considered defective. All defective Nozzle Plates shall be removed from the lot.

4.3.3.6 Piston Free Movement Test on Fin and Closure Assembly with the Fin Retainer and Contact Disc disengaged from the Fin Blades (see dwg. 9266959)- Major Defect - Code No. 03010. - The sampling plan for this test shall be in accordance with Standard MIL-STD-105 using Inspection Level II with an AQL 0.40 percent. This test shall be performed as specified in 4.4.6 using equipment in accordance with 4.3.4. Any Fin and Nozzle Assembly with the Fin Retainer and Contact Disc disengaged from the Fin Blades in which the Piston will not freely move one half (1/2) inch to the rear and return forward shall be considered defective.

NOTE: This test shall be performed in such a manner that the Piston Movement Restriction of Note 6 dwg. 9266959 is maintained.

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4.3.3.7 Fin Blade Free Fall Test on Fin and Closure Assembly with the Fin Retainer and Contact Disc disengaged from the Fin Blade (see 3.11)- Major Defect - Code No. 03011. - The sampling plan for this test shall be in accordance with Standard MIL-STD-105 using Inspection Level II with an AQL of 0.40 percent. This test shall be performed as specified in 4.4.7 using equipment in accordance with 4.3.4. Any Fin and Closure Assembly with the Fin Retainer and Contact Disc disengaged from the Fin Blades on which any Fin Blade does not fall of its own weight when tested shall be considered defective.

4.3.3.8 Fin Pivot Pin Push Test on Fin and Closure Assembly (see 3.5)- Major Defect - Code No. 03012. - The sampling plan for this test shall be in accordance with Standard MIL-STD-105 using Inspection Level II with an AQL 0.40 percent. This test shall be performed as specified in 4.4.8 using equipment in accordance with 4.3.4. Any Fin and Closure Assembly in which any Pin moves in relation to the fin when tested shall be considered defective. Any Fin and Closure Assembly in which any metal fatigue is in evidence on the Fin Blade shall be considered defective.

4.3.3.9 Live Lead and Ground Lead Pull Test on Fin and Closure Assembly, with the Fin Retainer and Contact Disc disengaged from the Fin Blades. (Dwg. 9266959)- Major Defect - Code No. 03013. - The sampling plan for this test shall be in accordance with Standard MIL-STD-105 using Inspection Level II with an AQL of 0.40 percent. This test shall be performed as specified in 4.4.9 using equipment in accordance with 4.3.4. If the Electrical Lead Live Assembly or the Ground Lead exhibit separation or relative movement between any of the compoments, the Fin and Closure Assembly shall be considered defective.

4.3.3.10 Insulation Integrity Test on Fin and Closure Assembly (see 3.6)- Major Defect - Code No. 03014. - The sampling plan for this test shall be in accordance with Standard MIL-STD-105 using Inspection Level II with an AQL of 0.25 percent. The test shall be performed as specified in 4.4.10 using equipment in accordance with 4.3.4. Any Fin and Closure Assembly which indicates less than the required resistance shall be considered defective.

4.3.3.11 Electrical Continuity Test on Live Lead of Fin and Closure Assembly - (see dwg. 9266959)- Major Defect - Code No. 03015. - The sampling plan for this test shall be in accordance

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with Standard MIL-STD-105 using Inspection Level II with an AQL of 0.25 percent. This test shall be performed as specified in 4.4.11 using equipment in accordance with 4.3.4. Any Fin and Closure Assembly, in which the resistance in the live lead is indicated as exceeding that specified in Note 10 on dwg. 9266959 shall be considered defective.

4.3.3.12 Nut Torque Test (see dwg. 9266959).-Major Defect - Code No. 03016. - The sampling plan for this test shall be in accordance with Standard MIL-STD-105 using Inspection Level II with an AQL of 0.40 percent. This test shall be as specified in 4.4.12 using equipment in accordance with 4.3.4. Any Fin and Closure Assembly that does not have the MS Nut torqued to a minimum of 12 inch pounds shall be considered defective.

4.3.3.13 Closure Assembly Alcohol Inspection (see 3.12).- Critical Defect - Code No. 16029. - The Closure Assembly shall be one hundred percent inspected and the test performed as specified in 4.4.13 using equipment in accordance with 4.3.4. Any Closure Assembly that does not meet the requirements, of 3.12 shall be considered defective.

4.3.3.14 Electrical Continuity Test on Ground Lead of Fin and Closure Assembly (see 3.13).-Major Defect - Code No. 03017. - The sampling plan for this test shall be in accordance with Standard MIL-STD-105 using Inspection Level II with an AQL of 0.25 percent. This test shall be performed as specified in 4.4.14 using equipment in accordance with 4.3.4. Any fin and closure assembly, in which the resistance in the ground circuit is indicated as exceeding that specified shall be considered defective.

4.3.3.15 Air Test on Closure Assembly (see 3.14).-Major Defect - Code No. 16030. One hundred (100) closure assemblies per inspection lot shall be subjected to this test. This test shall be performed as specified in 4.4.15 using equipment in accordance with 4.3.4. Any closure assembly which indicates a leakage rate in excess of the requirement shall be considered defective. One defective shall reject the lot.

4.3.3.16 Nozzle Plate Tensile Test.-Major Defect - Code No. 04056 (see dwg. 9266954). Each furnace (heat treat) lot of bar stock shall be sampled and tested for materiel strength. Furnace (heat treat) lots of bar stock shall be sampled and tested according to the following: Three (3) tensile test specimens from each furnace (heat treat) lot of bar stock shall be obtained and subjected to a tensile pull test as specified in 4.4.16 using equipment in accordance with 4.3.4. Failure of one or more specimens to meet the requirements of dwg. 9266954 shall cause furnace (heat treat) bar stock rejection. In the event that lot

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rejection criteria are reached due to no test (machine failure, specimen failing outside gage length, etc) each no test specimen shall be replaced in the sample by another specimen obtained from the same bar stock heat treat lot.

4.3.4 Inspection Equipment. - Equipment Tabulation Number ET 9266959 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government Rights to Documentation. - Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and wherever inspection equipment drawings are needed.

4.3.4.2 Supply and Maintenance. - Supply and maintenance of the equipment listed on the ET shall be in accordance with Specification MIL-I-45607.

4.4 Test Methods and Procedures

4.4.1 Fin Blade Tensile Test (see 4.3.3.1)

4.4.1.1 Fin Blade Material Verification Test (see 4.3.3.1.1) A specimen shall be prepared from each fin blade in the sample. The specimen shall be machined from a finished heat treated fin. The specimen shall be approximately 5.9 inches long with shorter grip length, but shall otherwise conform to Figure 6, Para. 4.3.1 ASTM-E8 Federal Test Method STD No. 151 with specific dimensions as follows:

W - 0.500 + .010	R - 0.500
C - 0.750 (max)	G - 2.000 + .005
A - 2.250 (min)	T - 0.100 (nominal)

Each specimen shall be pulled in accordance with tension test method ASTM-E8 of Federal Test Method STD. 151.

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4.4.1.2 Fin Blade Pin Pull Test (see 4.3.3.1.2). Fin Blades and Pins used for this Fin Blade Pin Pull Test shall be selected from lots that are to be utilized during production. The Fin Blade shall be heated to 190oF. maximum before installing the pin. Each Fin Blade with pin inserted in the sample shall be subjected to a pull force that is perpendicular to the Fin Blade Hole/Pin Centerline. The pin shall be supported on both sides of the Fin Blade with a distance between supports of .3740 + .0025. The aft portion of the Fin Blade shall be gripped using suitable gripping jaws which shall not grip any portion of the Fin Blade two inches or more from the aft end of the Fin Blade. The force shall be applied until failure of the fin blade occurs or until a load force of three thousand pounds (3000 lbs.) is reached. Observation shall be made of the load at failure or 3000 lbs. which ever condition is met.

4.4.2 Compression Test on Cross Head (see 4.3.3.2). - Each Cross Head in the sample shall be subjected to a compression load of one thousand (1,000) pounds, applied alternately at the outer ends of each set of diametrically opposite arms. Visual observation shall be made for permanent deformation of the Cross Head.

4.4.3 Bending Test on Cross Head (see 4.3.3.3). - Each Cross Head in the sample shall be subjected to a bending load of five hundred (500) pounds applied at the center of the Cross Head by a .217 inch, Maximum, diameter ram, with the Cross Head supported alternately by supports under each set of diametrically opposite arms. The supports shall be a minimum of .950 inches apart. Visual observation shall be made for permanent deformation of the Cross Head.

4.4.4 Pressure Test on Piston Assembly (see 4.3.3.4). - Each Piston Assembly in the sample shall be held in such a way as to subject that portion of the assembly forward of the "O" ring groove to two thousand (2000) psig pressure, plus fifty (50) psig. minus zero (0) psig for at least two (2) seconds. Observation shall be made for leakage or pressure drop.

4.4.5 Magnetic Particle Inspection of Nozzle Plate (see 4.3.3.5). - Each Nozzle Plate shall be magnetic particle inspected in accordance with MIL-I-6868.

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4.4.6 Piston Free Movement Test (see 4.3.3.6). - Each Fin and Closure Assembly in the sample with the Fin Retainer and Contact Disc disengaged from the Fin Blades shall be partially functioned by moving the Piston one half (1/2) inch aft. The piston shall then be returned to the forward position. Do not move the piston more than five eights (5/8) inch. Excessive movement will damage serrations on the piston.

4.4.7 Fin Blade Free Fall Test (see 4.3.3.7). - Each Fin and Closure Assembly in the sample, with the Fin Retainer and Contact Disc disengaged from the fin blades shall be supported in the detent groove of the nozzle plate. The fin blades shall be raised to a position which is at least sixty (60) degrees above the horizontal. The fins shall be released and allowed to fall of their own weight.

4.4.8 Fin Pivot Pin Push Test (see 4.3.3.8). - Each Fin and Closure Assembly in the sample shall be held with the Fin Blade positively positioned. A static load of twenty (20) pounds shall be applied to the end of each pin. Observation shall be made for movement between the Fin Blade and Pin during application of the load.

4.4.9 Lead Pull Test (see 4.3.3.9). - Each Fin and Closure in the sample with the Fin Retainer and Contact Disc disengaged from the Fin Blades shall undergo the following: The Nozzle Plate shall be held with the Cross Head secured from movement. Alternately, the forward end of the live lead, and the contact disc on the live lead shall have a twelve (12) pound pull applied for at least two (2) seconds. Observation shall be made for relative movement between any part of any lead and the nozzle plate. In addition, the ground lead and the soldered connection of the Ground Clip to the piston boss shall withstand a two (2) pound pull for at least two (2) seconds. Observation shall be made for relative movement between any part of the lead and the nozzle plate.

4.4.10 Insulation Integrity Test - (see 4.3.3.10). - Each Nozzle and Fin Assembly in the sample shall have a one hundred (100) volt potential applied to measure the resistance between the Contact Disc and the bourrelet of the Nozzle Plate.

4.4.11 Electrical Continuity Test (Live Lead) (see 4.3.3.11). The resistance of each Fin and Closure Assembly, in the sample shall be measured between the Contact Disc and the forward end of the Electrical Lead Live Assembly.

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4.4.12 Nut Torque Test (see 4.3.3.12). Each Fin and Closure Assembly in the sample, with the Fin Retainer and the Contact Disc disengaged from the Fin Blades shall be supported in the detent groove on the nozzle plate. The piston may be partially functioned by moving the piston one half (1/2) inch aft to expose the MS Nut. Do not move the piston more than five eighths (5/8) inch. A torque wrench shall be placed on the MS nut and rotated in a direction so as to tighten the nut with the piston held. Any MS Nut that moves, i.e., rotates or turns, when observation on the torque wrench indicates that the minimum torque requirement (12 inch pounds) has not been obtained, shall be considered defective.

4.4.13 Closure Assembly Alcohol Inspection (see 4.3.3.13). Each Closure Assembly shall be alcohol inspected using the following procedure:

4.4.13.1 Swab the molded surfaces of the closure assembly with alcohol conforming to Specification O-M-232, Grade A or C, or Isopropyl Alcohol, TT-I-735 Grade A or Butyl Alcohol, TT-B-846.

4.4.13.2 Scan the molded surfaces as the alcohol evaporates. The presence of cracks open to the surface will be indicated by a wet line, which is usually continuous, after the neighboring areas appear dry. Wide or deep cracks will remain wet longer and will show a wider indication than narrow or shallow cracks.

NOTE: Alcohol is highly flammable.

4.4.14 Electrical Continuity Test (Ground Lead) (see 4.3.3.14). The resistance of each fin and closure assembly ground lead in the sample shall be measured between the bourrelet on the nozzle plate and the forward end of the ground lead.

4.4.15 Air Test on Closure Assembly (see 4.3.3.15). Each closure assembly in the sample shall be subjected to ten (10) psig air pressure internally for at least ten (10) seconds. Observation shall be made for the leakage rate (3.14). During testing, the piston hole and the nozzles at the throat, but not to extend to the aft interface of the throat insert, are to be plugged.

4.4.16 Nozzle Plate Tensile Test (see 4.3.3.16). A specimen shall be prepared from each bar stock in the sample. Each specimen shall be prepared and pulled in accordance with test method ASTM-E8 of Federal Test Method STD. No. 151.

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5 PREPARATION FOR DELIVERY.

5.1 Packaging, Packing and Marking.

5.1.1 Level C. - Fin and Closure Assemblies shall be packaged, packed and marked in accordance with MIL-STD-1169 requirements and the following additional instructions: Assemblies shall be placed in a fiberboard box complete with half-slotted partitions assembled to form 30 cells in a 6 by 5 arrangement. Filler pads consisting of plain fiberboard sheets shall be added as required for a tight pack. The box shall be constructed in accordance with the requirements of Specification PPP-B-636 Type CF, Class Domestic, Variety SW, Grade 200 min. Box style shall be CSSC. The partitioned cells and filler pads shall be of the same test board as the box. Box shall be closed by taping or stapling.

6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data Cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection Code Numbers. - The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection Equipment. - The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection Equipment Lists. - Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraph 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.2 Inspection Equipment List Codes. - The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army Designs. - Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor Designs. - Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, naterials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000. Category F, Form 2, Legibility and reproductibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of Designs. - Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND9. All other

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designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of First article Samples. - Instruction as to the location for the evaluation of the First Article sample shall be obtained from the Contracting Officer. Upon receipt of such request, the Contracting Officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquires should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-NI.

Notice: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture use, or sell any patented invention that may in any way be related thereto.

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Project No. 1340

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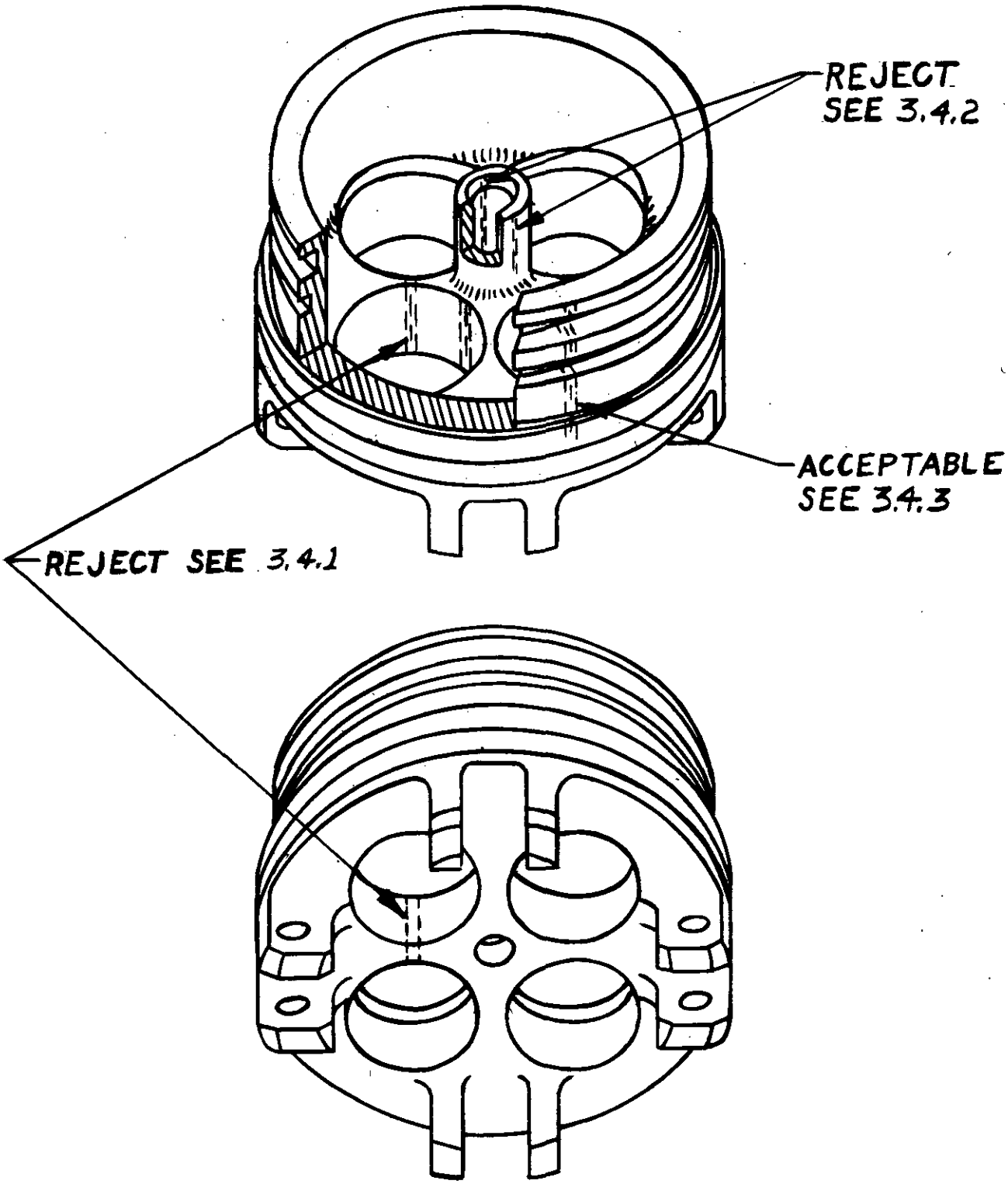


FIGURE 1.

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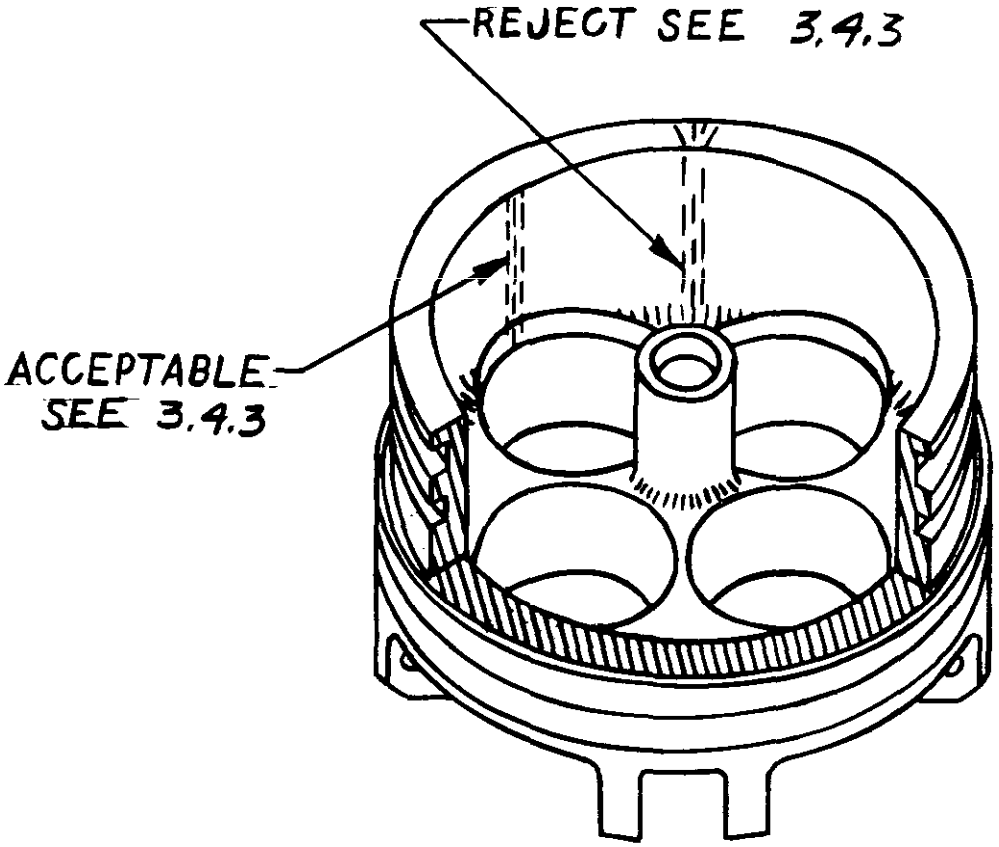


FIGURE 2

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

OFFICIAL BUSINESS

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DEPARTMENT OF THE ARMY

COMMANDING OFFICER
United States Army
Picatinny Arsenal
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Dover, New Jersey 07801

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